

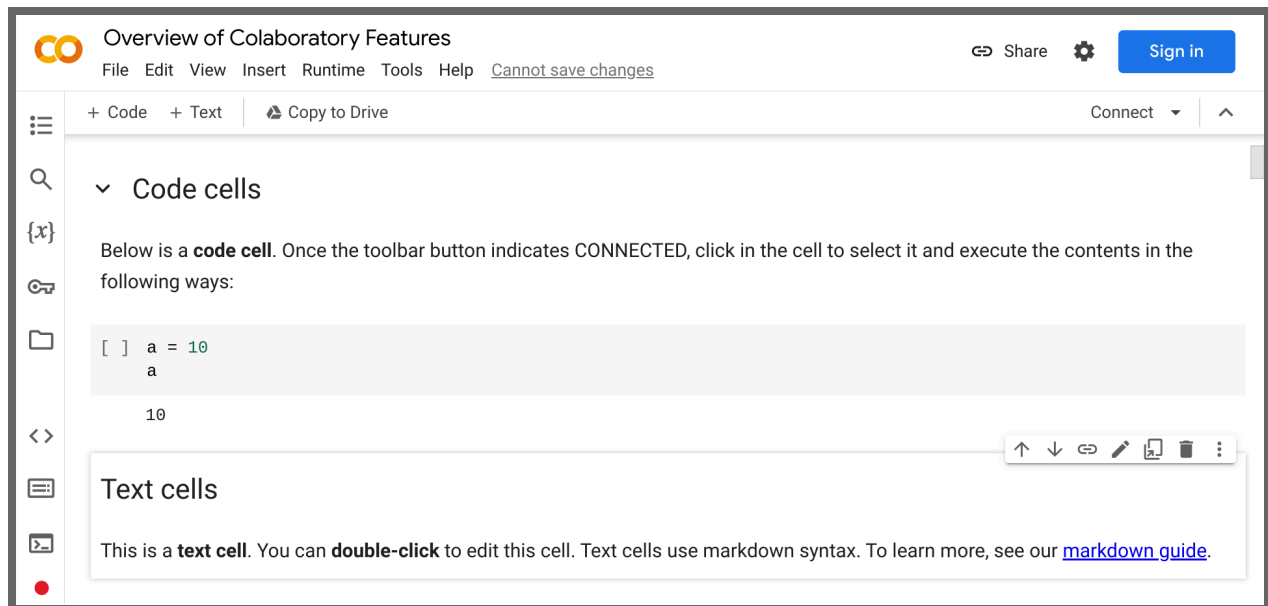
Managed notebooks

Previously, you were introduced to Jupyter notebooks as a way of collaborating on datasets with colleagues. Jupyter notebooks are open-source web applications that integrate live code, equations, and visualizations right in your notebook. Jupyter notebooks also allow you to share and access data from different sources in near real-time. Both Google Colaboratory (Colab) and Vertex AI Workbench are versions of Jupyter Notebooks that support data cleaning, data visualization, and machine learning. In this reading, you'll learn more about how Colab and Vertex AI Workbench can support your data analysis.

Google Colaboratory

Google Colaboratory (Colab) is a cloud-hosted version of a Jupyter Notebook that allows users to write and execute Python in a browser, with no configuration required. Colab notebooks are a series of cells. Each cell can contain either text or executable code. Cells that contain executable code also contain the output once the code is run.

This graphic details the difference between code cells and text cells in Colab:

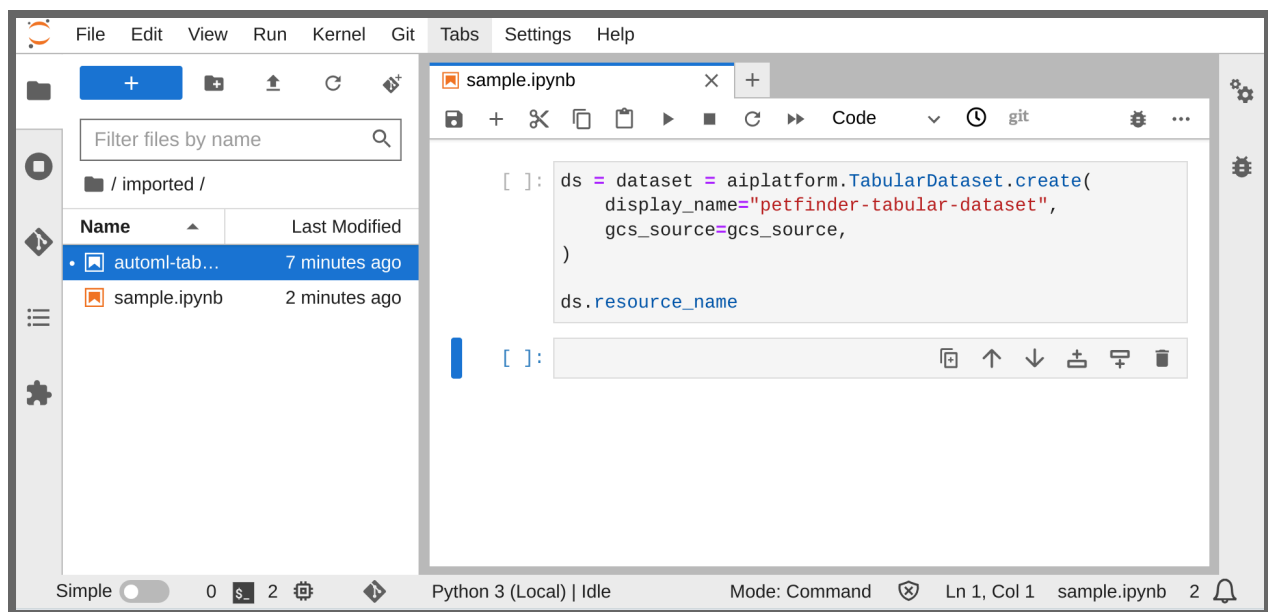


Because Colab is a Jupyter Notebook service, you can write and execute Python in code cells. You can move both text and code cells by selecting and dragging the cells. As a shareable notebook, anyone with editing access can update the notebook for all users.

Vertex AI Workbench

Another version of a Jupyter Notebook is Vertex AI Workbench. This service exists within the Google Cloud platform, and allows you to access and explore data from within a Jupyter notebook. Vertex AI Workbench also allows you to use BigQuery to query data stored in Google Cloud Storage from your notebook. It offers the ability to run your entire data science workflow from within your notebook.

This graphic is an example of code written in cells in Vertex AI Workbench:



Vertex AI Workbench allows you to automate updates to your data models using code written in your notebook. Integration with all of Google Cloud's services provides the opportunity to maintain your entire data pipeline via your collaborative notebook.

Colab versus Vertex AI Workbench

Since Colab and Vertex AI Workbench are both Jupyter Notebooks, selecting which to use will depend on your project needs and the type of collaboration you require. This table provides an overview of some of the differences between Colab and Vertex AI Workbench:

Notebook	Use	Cost	Hardware and usage	Notes
Colab	- Individuals - Educational settings	None	- Limited hardware - Limited usage - Cloud-based	Commonly used for code sharing
Vertex AI Workbench	- Large organizations - Enterprises	Paid service	- Control over hardware size and scale - Control over environment setting - Cloud-based	- Includes enterprise-grade security - Includes cross-product integration

Detail/Description	Colab	Vertex AI Workbench
Use	• Individuals • Educational settings	• Larger organizations • Enterprise
Cost	• No cost	• Paid service
Hardware and usage	• Limited hardware • Limited usage • Cloud-based	• Control over hardware size and scale • Control over environment setting • Cloud-based
Additional notes	• Commonly used for code sharing	• Includes enterprise-grade security • Includes cross-product integration

Key takeaways

Jupyter-based notebooks provide opportunities for collaboration and management of the entire data science pipeline. Google-based options also include the ability to integrate other Google Cloud features into your notebook, allowing for a seamless integration of data

management tools. Both Colab and Vertex AI Workbench provide the option to share data and code in real-time. And, they each offer specialized features based on your use case.

Resources for more information

To explore notebooks on your own, check out these links:

- Welcome to Colab provides an overview of the basic features of Colab along with examples of notebooks: <https://colab.sandbox.google.com/#scrollTo=ISrWNR3MuFUS>
- Google Cloud provides introductory tutorials to Vertex AI Workbench: <https://cloud.google.com/vertex-ai/docs/workbench/notebooks>